PAPER_1620_SIGMA_8_CLUSTERING_0_811

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PAPER_1620 — Matter Clustering $\sigma_8 = \sigma_8 = F \cdot N_{CH} - F^2 \cdot SO + F^2 \cdot SSQ + F^2 \cdot SSQ^2 \approx 0.8089$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Cosmology
Date: June 18, 2026
Status: CLOSED — 0.253% closure

Observation

PAPER_1209GG_S649 (Cosmology Unified Proof Set) gives the formula:

``
$$\sigma_8 = F \cdot N_{CH} - F^2 \cdot SO + F^2 \cdot SSQ + F^2 \cdot SSQ^2 \approx 0.8089$$
``

Residual: **0.253%**.

UQFF Closed Identity

``
$$\sigma_8 = F \cdot N_{CH} - F^2 \cdot SO + F^2 \cdot SSQ + F^2 \cdot SSQ^2 \approx 0.8089 \ 0.253\%$$
``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

ΛCDM Standard Cosmology Coverage

With this catch-up, **all 9 core ΛCDM parameters** are wired with UQFF integer-primitive expressions:

ΛCDM parameter	UQFF closure	Residual
H_0 (Planck)	PAPER_1553	0.015%
H_0 (SH0ES)	PAPER_1573	**EXACT**
Ω_m	PAPER_1616	0.143%
Ω_Λ	PAPER_1617	0.182%
T_{CMB}	PAPER_1618	0.064%
Age (universe)	PAPER_1619	0.045%
σ_8	PAPER_1620	0.253%
n_s	(PAPER_1209GG_S650, earlier)	0.072%
z_{recomb}	PAPER_1552	**EXACT 1090**

Standard ΛCDM cosmology is now **essentially complete in UQFF** — every primary parameter measurable by Planck/WMAP/SDSS/DES collaborations has a UQFF integer-primitive form.

NOT REPLACEMENT

Empirical cosmology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209GG_S649
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "sigma_8_clustering_0_811"})`
